PROJECT CHARTER

U.S. Federal Medicaid Drug Utilization & Cost Intelligence

A 10-Year State-Level Analysis (2016–2025)

Project Name	Medicaid Drug Spend Intelligence & Cost Optimization	Version 3.0
Prepared By	Bhumi Nagar — Business Analyst	April 2026
Project Type	Healthcare Business Analysis | Payer Analytics | Cost & Utilization Intelligence	
Primary Stakeholder	Medicaid Managed Care Organization (MCO) & Payer Analytics Team	
Data Source	CMS Medicaid State Drug Utilization Data — 2016 to 2025 (Medicaid.gov)	
Tools & Technologies	PostgreSQL (Data Engineering & KPI Design) | Power BI (Dashboard & Visualization)	

1. BUSINESS PROBLEM STATEMENT

Medicaid managed care organizations (MCOs) face a critical and growing challenge: pharmaceutical spending continues to rise year-over-year, yet most payer analytics teams lack the structured visibility to understand what is actually driving those costs.

Without a clear, data-driven picture of drug utilization patterns, cost concentration, and state-level variation, MCO decision-makers are forced to make formulary management, rebate negotiation, and cost containment decisions without adequate analytical support.

Specific gaps this project addresses:

- No visibility into which drugs are driving the majority of pharmaceutical spend
- No structured understanding of how spending has trended over a 10-year period
- No state-level benchmarking to identify geographic outliers and high-cost markets
- No analysis of whether cost increases are driven by price inflation or utilization volume
- No comparison of spend efficiency between Managed Care (MCOU) and Fee-for-Service (FFSU) models

2. PROJECT OBJECTIVES

This project was designed to deliver the following outcomes for the MCO payer analytics team:

1	**Identify top cost-driving drugs —** determine which drugs account for the majority of Medicaid pharmaceutical spend using Pareto analysis
2	**Map 10-year spending trends —** analyze year-over-year growth in spend, prescriptions, and units to identify inflection points and demand patterns

3	**Benchmark state-level performance —** compare pharmaceutical spending and cost-per-prescription across 50+ states to identify geographic outliers
4	**Decompose cost drivers —** separate price-driven cost increases from volume-driven increases to support targeted intervention strategy
5	**Compare MCOU vs FFSU efficiency —** evaluate which utilization model delivers better cost-per-prescription outcomes
6	**Deliver actionable recommendations —** translate all findings into business-ready cost optimization recommendations the MCO analytics team can act on directly

3. PROJECT SCOPE

In Scope:

- 10 years of CMS Medicaid State Drug Utilization data (2016–2025)
- All 50 U.S. states + District of Columbia
- Both utilization types: Managed Care (MCOU) and Fee-for-Service (FFSU)
- Drug-level, state-level, year-level, and utilization-type analysis
- KPI design, data pipeline development, and dashboard delivery
- Business recommendations for cost optimization and formulary management

Out of Scope:

- Patient-level or beneficiary-level data
- Clinical efficacy or therapeutic outcome analysis
- Real-time or live data feeds — analysis based on historical CMS published data
- Predictive modeling or forecasting
- Provider-level patient analysis
- Production deployment infrastructure

4. STAKEHOLDER ANALYSIS

Stakeholder	Role	Interest	Influence
MCO Payer Analytics Team	Primary Consumer of Insights	Cost reduction, spend visibility, formulary management	**High**
Medicaid Program Director	Policy & Budget Oversight	Program performance, compliance, cost containment	**High**
Pharma Market Access Teams	Secondary Consumer	Drug positioning, formulary access strategy	**Medium**
CMS / Federal Policy Analysts	Data Owner / Regulatory Body	Policy impact, spend accountability	**Medium**
Finance & Operations Teams	Budget Planning	Cost monitoring, spend forecasting	**Medium**

5. SUCCESS METRICS

This project will be considered successful when the following outcomes are delivered:

Success Criterion	Measurement
Cost concentration identified	Pareto analysis showing top drugs contributing to 80% of spend
10-year trend analysis delivered	10-year YoY spend, prescription, and unit trends visualized across 2016-2024
State benchmarking complete	All 50+ states ranked by spend and cost per prescription
Cost drivers decomposed	Price vs. volume analysis completed and documented, and visualized
MCOU vs FFSU comparison done	Side-by-side efficiency comparison with clear findings documented
Actionable recommendations delivered	Minimum 5 business recommendations with supporting data evidence
Executive dashboard delivered	Multi-page Power BI dashboard accessible and interpretable by non-technical stakeholders

6. ASSUMPTIONS & CONSTRAINTS

Assumptions:

- CMS published Medicaid data is accurate and representative of actual state drug utilization
- Suppressed data rows indicate low-volume records withheld for privacy — treated as NULL in analysis
- MCOU (Managed Care) and FFSU (Fee-for-Service) are the two primary utilization models for comparison
- Drug product names in the dataset are sufficiently standardized for product-level aggregation
- 2025 data excluded from trend visuals due to partial-year reporting – noted in documentation

Constraints:

- Analysis limited to publicly available CMS aggregated data — no patient-level or beneficiary identifiers
- Drug therapeutic classification not available in source data — analysis at product name level only
- Invalid geographic values ('XX') excluded from state-level analysis — documented as data quality consideration
- No real-time data integration — findings reflect historical patterns based on published CMS datasets

7. RISKS & MITIGATION

Risk	Impact	Mitigation
Partial-year 2025 data	Misleading trend	2025 excluded from executive trend visuals — documented in data quality notes
Suppressed data records	Incomplete calculations	Suppression handled via conditional logic — NULLs applied; tracked via separate suppression KPI views

Invalid geographic labels ('XX')	Distorted state analysis	'XX' records excluded from geographic visuals — documented as unclassified jurisdictions
Decimal overflow in Power BI import	Data import instability	Isolated to specific high-precision records — managed through controlled data type formatting and load sequencing

8. PROJECT DELIVERABLES

#	Deliverable	Description	Status
1	**Project Charter**	Project scope, objectives, stakeholders, success metrics	☑ Complete
2	**Business Requirements Document (BRD)**	Stakeholder needs, business questions, functional requirements, KPI definitions	☑ Complete
3	**Data Pipeline (SQL)**	Raw → Staging → Analytics schema with 11 KPI views in PostgreSQL	☑ Complete
4	**Process Flow Diagrams**	As-Is / To-Be process maps and data architecture flow (Lucidchart)	☑ Complete
5	**KPI Definition Document**	Definitions, formulas, and business purpose for each KPI metric	☑ Complete
6	**Power BI Dashboard**	6-page multi-dimensional dashboard for executive and analytical use	☑ Complete
7	**Executive Summary**	1-page business-ready findings and recommendations document	☑ Complete

9. ANALYST SIGN-OFF

This Project Charter has been prepared by Bhumi Nagar and represents the foundational scope, objectives, stakeholder alignment, and success criteria for the Medicaid Drug Utilization & Cost Intelligence project. All subsequent deliverables — BRD, KPI definitions, process maps, Power BI dashboard, and executive summary — will be developed in alignment with the objectives and constraints defined in this document.

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